

Developing and maintaining a policy-focussed, comprehensive, energy systems model in a developing country context: The case of SATIM in South Africa

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ABSTRACT

South Africa faces a myriad of energy-related challenges, including severe constraints in electricity supply, a rising cost of electricity to consumers, high levels of energy poverty, and a reliance on coal in both energy supply and demand sectors. These challenges exist in a context of significant development imperatives and commitments to decarbonise. This paper presents the South African TIMES model (SATIM), a comprehensive energy systems model developed and maintained over two decades, which has proved useful in informing energy and climate policy in South Africa. We explore how SATIM has been adapted to overcome data limitations and address unique and complex policy, development and broader socio-economic challenges faced by developing countries. The model's evolution, structure, and applications are discussed, highlighting its role in providing evidence-based insights for critical policy decisions. We argue that SATIM's long-term development, flexibility, and integration with other models offer valuable lessons for energy systems modelling in developing country contexts.

1. Introduction

Energy systems models are essential tools for evidence-based policy-making, particularly in addressing challenges such as climate change mitigation and sustainable development. Used independently or in conjunction with other kinds of models, they are now an essential component in the development of an appropriate evidence base for making energy policy. It is very difficult to consider key policy options to meet challenges such as climate change mitigation without a systematic approach which is a) capable of quantifying key policy-relevant indicators such as economy-wide GHG emissions, and b) assessing the direct and indirect impact of proposed policies and measures across the economy expressed in indicators which policy makers consider paramount. Moreover, as discussed below, as an outcome of both international climate policy and dramatic changes in energy economics and technology on both the supply and demand sides, a 'business as usual' approach to policy making based on incremental change (and no significant structural changes) no longer has any utility. Energy policy options are now clearly required not only to meet traditional objectives

such as energy security, and more recent policy objectives such as decarbonisation, but must also address other key sustainable development objectives such as reducing poverty and inequality, reducing other environmental impacts such as air pollution, and reducing unemployment, directly or indirectly, and supporting economic development.

These policy challenges are far more acute in developing countries such as South Africa, which face the triple challenge of poverty, unemployment and inequality, far more complex energy system at household and community level, and which also have highly carbon-intensive energy systems that are already transitioning. Moreover, there are additional challenges in undertaking model-based policy analysis. These lie in developing national models (including data challenges), undertaking appropriate modelling analyses, and sustaining both the models and the relevant expertise over time. Complex systems models are social enterprises in that these rest on a set of extraneous institutional arrangements which both provide the relevant data and the context for their use, and more importantly, are the product of a concentration of diverse expertise.

The effective use of complex systems models for policy-making requires several conditions to be met. First, models must be technically

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List of acronyms

AFOLU - Agriculture, Forestry and Other Land Use	NZ - Net Zero
BRT - Bus Rapid Transit	OCGT - Open Cycle Gas Turbine
CBECS - Commercial Buildings Energy Consumption Survey	OSeMOSYS - Open Source Energy Modelling System
CCDR - Country Climate and Development Report	PCC - Presidential Climate Commission
CCGT - Combined Cycle Gas Turbine	PEM - Polymer Electrolyte Membrane
CGE - Computable General Equilibrium	pkm - passenger kilometres
CO ₂ - Carbon Dioxide	PLEXOS - (Not an acronym, but a proprietary software name)
CSIR - Council for Scientific and Industrial Research	PM10 - Particulate matter with a diameter of 10 µm or less
CTL - Coal-to-Liquids	PV - Photovoltaic
DFFE - Department of Forestry, Fisheries and Environment	RE - Renewable Energy
DMRE - Department of Minerals Resources and Energy	SA - South Africa
EB - Energy Balance	SAGE - South African Computable General Equilibrium
eSAGE - energy-enhanced South African General Equilibrium	SANS - South African National Standard
ESRG - Energy Systems Research Group	SAPP - Southern African Power Pool
GTL - Gas-to-Liquids	SATIM - South African TIMES Model
GDP - Gross Domestic Product	SATIMGE - South African TIMES Model linked to an economy-wide model
GHG - Greenhouse Gas	SET - Sectoral Emissions Target
HFO - Heavy Fuel Oil	SMR - Steam Methane Reforming
HVAC - Heating, Ventilation, and Air Conditioning	SUT - Supply and Use Tables (StatsSA)
IPCC - Intergovernmental Panel on Climate Change	SO _x - Sulphur oxide
IRP - Integrated Resource Plan	SSA - Sub Saharan Africa
JET - Just Energy Transition	TFC - Total Final Consumption
JET IP - Just Energy Transition Investment Plan	tkm - tonne kilometres
LPG - Liquid Petroleum Gas	TIMES - The Integrated MARKAL-EFOM System
LTMS - Long Term Mitigation Scenarios	UCT - University of Cape Town
MARKAL - Market Allocation	UE - Useful Energy
NDC - Nationally Determined Contribution	UNEP - United Nations Environment Programme
NO _x - Nitrogen Oxide	UNFCCC - United Nations Framework Convention on Climate Change

credible; in other words, the structure and data comprising the model must be technically appropriate to address the relevant policy problem. Second, the modelling process must be accepted as legitimate by at least some key stakeholders¹ – stakeholders must both accept the technical credibility of the model and trust that modellers are not themselves political actors who will exploit the ‘black box’ nature of many models to achieve their own or other stakeholders’ political goals. Finally, modelling analyses must be underpinned by appropriate expertise in developing and maintaining models, designing modelling analyses appropriate to the relevant policy problem, and interpretation of the results/identifying implications for policies and measures of the analyses. This last condition includes the means to reproduce this capacity through time.

Developing, maintaining and using such models effectively in developing countries presents unique challenges, including data limitations, complex socio-economic factors, and rapidly changing policy landscapes. This paper considers, as a case study, the evolution, structure, policy relevance and maintenance over time of the South African TIMES model, and aims to derive lessons which have broader relevance for the challenges of developing and maintaining policy-relevant energy systems models in developing country contexts, given data limitations and complex socio-economic policy challenges.

We present the case of the South African TIMES model (SATIM), an economy-wide energy systems model that has been developed and maintained for over 14 years. SATIM has evolved to address South Africa’s specific energy policy challenges, including high levels of poverty, unemployment, and inequality, a carbon-intensive economy,

and an ongoing electricity sector crisis.

The paper is structured in three parts. Firstly, the context in which energy systems modelling takes place is presented through a review of South Africa’s energy policy domain (Section 2), and energy systems models that have been used in South Africa to inform policy (Section 3). Secondly, a detailed overview of the structure and development of SATIM is presented in Section 4. Finally, the paper presents applications of SATIM to inform policy in South Africa, along with a discussion of current limitations of the model (Section 5), and the focus and motivation for future improvements to the model (discussed in Section 6). Conclusions can be found in Section 7.

2. The South African energy policy domain

2.1. Historical context and development challenges

South Africa is a middle-income developing country with critical development challenges, including unusually high levels of poverty, unemployment and inequality, a massive skills deficit, a very carbon-intensive coal-based economy, and an ongoing crisis in its electricity sector (since 2008). A complex historical legacy of apartheid and colonialism bequeathed the newly democratic post-apartheid state with an economy which was built around mining and minerals processing, which in turn was based on coal-derived energy, and a lack of civic infrastructure. Since 1994, when the post-apartheid era began and the first democratically elected government took office, the population has grown by 50 % [1], and heavy industry and mining/minerals processing contribute a smaller share of GDP [2]. The post-apartheid economy grew at an average rate of 3.6 % in the period from 1994 to 2008, but since the

¹ Model-based analyses have contributed to both national consensus-building [16] as well as to resource advocacy coalitions in policy contestations – for instance on the speed and extent of decarbonisation.

global financial crisis annual growth has not exceeded 3 % and has largely been lower than 2 % [3].² The COVID pandemic, in common with other countries, resulted in weakening of the economy, with a contraction of 6 % reported in 2020 [3]. The economy has only recently reached its 2019 size. Within this context, a couple of key dilemmas face energy policy makers. First, infrastructure in both electricity and liquid fuels supply is ageing and will require massive investment in the coming decades; second, the carbon-intensive nature of the energy system will necessitate a shift from the current system to a decarbonised system, and third, addressing both these challenges will require consideration of a number of other policy objectives, including energy access and affordability, industrial policy and job creation, and other environmental objectives such as reducing air pollution and water use.

2.2. Evolution of the energy sector

The key features of South Africa's energy system were developed in the 1970s during the apartheid era and are almost entirely focused on coal. In the 1970s, the national transmission grid was completed, facilitating the construction of a fleet of massive mine-mouth coal plants. In addition to this, a coal-to-liquids (CTL) plant was constructed to minimise the energy security risk of oil sanctions imposed by the UN on South Africa, and coal is also the dominant fuel in industry to provide process heat. While the energy system has begun to diversify away from coal, in 2017 the country's energy balance indicates that coal still predominates, as presented in Fig. 1 below.

South Africa consumes around 6.2 EJ of primary energy [4,5] 85 % of which is derived from domestically supplied coal. Locally mined coal supplies around 80 % of electricity and around a fifth of liquid fuels through a CTL plant. Coal is also widely used for thermal energy in industry and for reduction in the metals sector. Energy imports are primarily crude oil and refined petroleum products. Liquid fuels used for transport are around 40 % of Total Fuel Consumption (TFC). Petrol and diesel consumption for road vehicles are the primary fuels used in the Transport sector accounting for approximately 80 % of the sector's total energy consumption. Industry consumes around 44 % of TFC, using around 8 % of coal and 62 % of electricity. The residential sector, and commercial sector use around 15 % of TFC. Commercial sector energy consumption primarily consists of electricity, which comprises almost 90 % of total energy use. Household electrification is currently around 85 % and whilst electricity provides the bulk of energy services to households, fuel use is diverse and includes the use of both coal and wood fuel, primarily in low-income rural or peri-urban households. Household energy poverty has been estimated at over 50 % [6]. Poor households face challenges in both energy access and affordability, and additional hazards from indoor and local air pollution.

2.3. Energy system transition and future challenges

On both the supply and demand side, the South African energy system is shifting away from its historical structure. Primary energy supply has been dominated by coal and liquid fuels, and energy demand, which has grown steadily and predictably from the 1970s, has departed from this historical trend. Electricity demand has been static for more than a decade, despite economic growth. The South African coal plant fleet will retire over the next four decades, and by 2050 only two of the current 15 coal plants will still be operational. These may also be retired earlier, for either economic reasons or to meet climate mitigation goals. Liquid fuels infrastructure, consisting of four refineries and a synthetic fuels plant, is ageing, and two of South Africa's largest refineries were retired in 2022. At the same time, South Africa has some of the best renewable energy (RE) resources in the world, and the country is seeing rapid investment in both utility-scale and on-site RE, facilitated by recent electricity

market reforms. The energy system thus faces a rapid departure from 'business as usual' in both supply and demand, and policy makers now confront a complex set of possible energy futures for the country. Each of these options would commit to developing infrastructure on a very large scale, with a significant degree of path dependency.

South Africa thus has several complex policy dilemmas intersecting with the energy system, for both energy and climate mitigation policy. South Africa has ratified the Paris Agreement, and its updated NDC communicated to the UNFCCC in 2021 commits the country to make substantial GHG emissions reductions by 2030. The much-feted Just Energy Transition Partnership (JETP), announced at COP26 in Glasgow, is focused on providing international support for the decarbonisation of the South African energy system, but there are ongoing debates amongst key stakeholders on how and when this should be achieved. There are closely related debates on the preferred energy future for the country, and not reaching a resolution will have disastrous economic consequences for the country.

Providing a robust evidence base to support decision makers in addressing these complex policy dilemmas, and the associated synergies and trade-offs, requires, in addition to other forms of analysis, the kind of complex systems analysis which full-sector energy and economic models can provide. Such analyses should be able to explore both the key dynamics in the energy and economic systems (and including GHG emissions and other environmental factors) and relate these dynamics to broader social and economic imperatives. Major national policy challenges currently facing South Africa in 2024 include the development and finalisation of a new Integrated Resource Plan for the electricity sector, the development of an Integrated Energy Plan, determining a new NDC mitigation target for 2035 (to be submitted to the UNFCCC early in 2025), determining Sectoral Emissions Targets (SETs) and carbon budgets for large emitters in terms of the newly-legislated Climate Change Act, determining a revised (and more ambitious) carbon tax level, revising South Africa's long-term low emissions development path (for submission to the UNFCCC in 2025) and the implementation of a competitive electricity market. Linked to these are numerous implementation challenges including unprecedented expansion of the national transmission grid, completion of the unbundling of Eskom and procuring the necessary investments in electricity supply, including in the context of the JET Investment Plan, which provide additional constraints.

3. Energy systems modelling in the South African context

South Africa's multi-faceted energy policy challenges necessitate an integrated approach to energy policy-making and planning. Whilst many different energy systems models, utilising a range of different methodologies, are in use globally across multiple regions to contribute to the necessary evidence base for policy-making in this context [7–11], energy optimisation models, which seek to maximise welfare by minimising total discounted system cost, are uniquely suited to analysing medium and long-term energy policy problems. Such analyses provide system-wide insights into the energy system in economic, social and environmental terms, and the complex synergies and trade-offs between each of these dimensions when providing for future energy needs. As such these are widely used tools globally for evaluating decisions that have long-term implications for the energy system under analysis [10, 12]. Notable amongst the optimisation modelling platforms that have been used in South Africa are PLEXOS, OSeMOSYS, MARKAL and TIMES.

Each of these platforms is suited to a specific set of applications. PLEXOS has been used by Eskom for power systems planning, including undertaking the technical work for South Africa's Integrated Resource Plans since 2010, and the CSIR's Energy Centre has used PLEXOS to provide an independent assessment of the Integrated Resource Plan (IRP) [13] as well as modelling long-term net zero pathways for the Presidential Climate Commission [14]. A full sector OSeMOSYS model

² Data are based on constant 2015 prices, expressed in U.S. dollars.

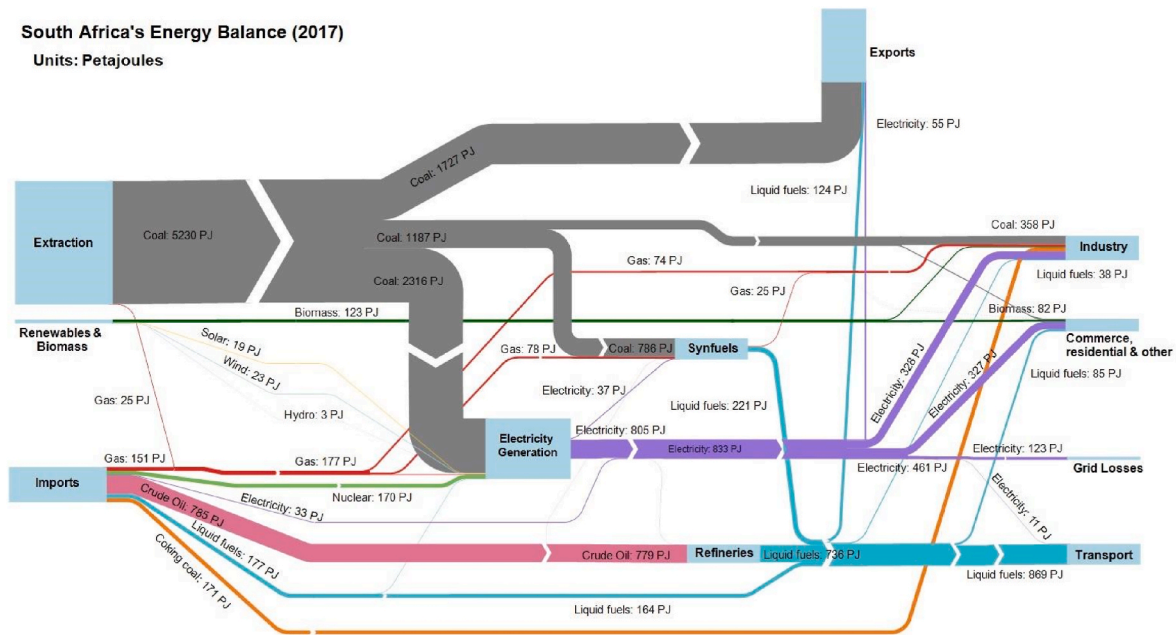


Fig. 1. Sankey diagram depicting the South African Energy Balance in 2017 (Source: ESRG).

was developed by the Department of Mineral Resources and Energy (DMRE) and was used to develop a draft Integrated Energy Plan in 2015 [15]. Whilst both the demand and supply side were represented in this model both have limited disaggregation and therefore limited ability to explore many of the pressing policy challenges that South Africa is facing.

A full sector MARKAL model of South Africa, was applied to the analysis of sustainable development goals and climate change mitigation options, for example in the development of the Long Term Mitigation Scenarios (LTMS) [16]. In the MARKAL model the demand and supply sectors were represented at a high level of detail, however there was limited ability to represent electricity demand and supply profiles at a sufficiently high level of detail, in particular as RE generation capacity increases.

All these modelling analyses face the same challenges, particularly a lack of data, and more importantly, the challenge of maintaining and building modelling capacity over time. Despite extensive efforts, only two modelling teams have persisted over the last two decades – the Energy Systems Research Group at the University of Cape Town, and the Eskom planning team, and only one focused post-graduate university course on energy analysis was available during this period.³ There are several reasons for this, and two are particularly important. First, in all contexts except Eskom, modelling groups are self-funded, and funding is inevitably project-based and for a duration of 6 months to 2 years. To maintain a modelling group through time, a consecutive sequence of projects need to be undertaken, and retaining skills on this short-term basis is challenging. Second, without continuity, each project requires a large portion of project time devoted to developing the relevant model.

SATIM was developed from the MARKAL model used for South Africa's Long Term Mitigation Scenarios Project and has been in use since 2010. The model is maintained and developed via a cycle of project-based funding, and the maintenance and continual enhancement of the model has had several key advantages, including being able to shift model development time to enhancing rather than building a new model, maintaining skills in specific sectors, and having the capability to respond to short-term requests from key stakeholders. In addition,

methodologies can be developed and enhanced to integrate other models into specific analyses over time.

SATIM is currently unique in its representation of the South African energy system – supply and demand sectors are represented in enough technical detail to explicitly capture the impact of demand side changes such as structural changes in the economy, industrial process and technology changes, fuel, technology and mode-switching in the transport sector, and efficiency gains in each sector. The structure and detail in the economic sectors (described in the following section) is designed to shed light on key policy priorities. SATIM is reviewed, updated and expanded regularly by the Energy Systems Research Group (ESRG) at the University of Cape Town (UCT). SATIM has also been hard linked to a CGE model, a development that has proved essential to ensure balanced development of the demand and supply sectors over time as well as allowing reporting on key development indicators such as labour growth, localisation, energy poverty and the JET for each scenario analysed. This is described further in sections 6.

4. SATIM: structure and development

4.1. Overview of SATIM

SATIM is structured to capture all extraction, imports, and exports, conversion of energy commodities and energy service demands across the economy. Each sector is disaggregated into sub-sectors which are structured to capture key elements that define each subsector. This section provides a detailed overview of the sectors represented in SATIM along with insights into the reasons behind their structure and parameterisation.

Base year fuel use across sectors is calibrated to an Energy Balance (EB), developed for SATIM, which replicates South Africa's energy system in 2017. This calibration is necessary as the official EB of South Africa, released periodically by the DMRE contains both insufficient sectoral detail as well as several anomalies and inconsistencies in sectoral consumption over the years (for example the large use of diesel attributed to the commercial sector and the large use of coal attributed to households). The EB calibration begins with the official energy balance for South Africa published by the DMRE [17], followed by a detailed review of these data and an expansion of sectoral detail using data available in other published sources (for instance to include the

³ The course was discontinued during a restructuring process, but there are plans to revive it.

ferrochrome and iron and steel industries as subsectors, and household disaggregation by income group). The primary sources of data for the calibration are outlined in Table A-1 in Appendix A. The base year is 2017 as it is the most recent year which can be considered a ‘normal’ year with no outlier effects from Covid-19 restrictions and the prevailing electricity supply constraints and widespread and worsening loadshedding that have occurred since 2020.

SATIM uses intra-year disaggregation into timeslices to model typical energy service load profiles for electricity use and RE resource availability. The timeslices capture diurnal and seasonal fluctuations. The temporal resolution is flexible and can be adjusted according to the research question and the available time⁴.

As a reflection of South Africa’s EB, SATIM considers both supply and demand sectors. Supply sectors include the extraction and trade of primary commodities, and the transformation sectors. Domestic resources for extraction include coal, oil, gas, and biomass. Existing oil and gas fields are largely depleted and there is no expansion of these allowed in the model. New gas fields are only included in specific scenarios, as extraction costs are highly uncertain. Biomass resources include Bagasse (to the sugar industry), Biomass for Pulp (to Pulp and Paper), Wood (to bio feedstock production, households and industry), and Other Biomass which includes biogas to the electricity sector. Due to South Africa’s historically large coal resources and resultant dependency on coal for power generation, SATIM has a detailed representation of South Africa’s coal sector.

- Coal resources are represented in two calorific grades, matching the South African coal profile (low-grade for power plants, some industrial sectors, and the CTL plant; and high-grade for other sectors and exports).
- Coal supply to the power sector is disaggregated per plant allowing supply to each power plant to be modelled separately at a specified price, which includes transport costs.

Transformation sectors include electricity generation, refineries, synthetic fuels plants (CTL and GTL), coke production, bio-feedstock and hydrogen production. The power sector in SATIM is structured into generation, transmission and distribution. By including both transmission and distribution as technologies, losses and infrastructure needs over these systems can be accounted for separately. This allows costs to be captured more accurately as demand grows, and the differences in distribution losses that are seen at the sector level to be included in the model.

Currently operating coal-fired power plants, South Africa’s only nuclear plant and gas turbines are represented as individual facilities, and the technology characteristics, water needs, and year of decommissioning are specified for each power plant. Model results thus give insight into the impact of different retirement schedules, performance and other characteristics such as local air pollution. The current retirement schedule (as per the Eskom Just Energy Transition (JET) strategy [18] indicates that 27 % of existing coal capacity will be retired by 2030 and 77 % by 2050; recent announcements indicate that the SA government may delay the retirement of some older plants.

New build options include potential imports from SAPP (by project) and nuclear, wind, solar PV (grid and rooftop), solar thermal, coal, diesel and gas (OCGT and CCGT), hydro and pumped storage plants as well as battery technologies. In addition to technology and cost parameters, all new build options are subject to constraints which limit either the cumulative capacity, annual maximum investments in new capacity, or the

growth in new capacity additions. This is done to reflect physical limitations such as port limitations, local skills capacity and manufacturing capacity.

Hydrogen, which has been earmarked as a key area for potential growth in South Africa [36], has been introduced into the model to supply both local demand and an export market. Further description of the SATIM supply sectors can be found in Appendix A.

On the demand side, demand for useful energy services (UE) is specified exogenously for each sub-sector. This allows fuel substitution and technology improvements to influence the efficiency at which energy services are met, and for the model to endogenously determine the final energy demand in each sector. Table 1 summarises the economic demand sectors in terms of their sub-sectors, end uses and sector drivers.

UE demand projections are based on the growth of primary drivers that govern the demand for energy services in each sector. GDP is the key primary driver and influences the demand for energy services in all sectors. Population growth directly impacts the demand for energy services in the residential and transport sectors and indirectly in all other demand sectors. GDP growth scenarios are derived from SAGE, which is in turn based on a social accounting matrix which guarantees consistency between sectors/sub-sectors growth rates which require goods and services at a commensurate rate from elsewhere in the economy. Further description of the methodology followed to characterise the demand sub-sectors within SATIM can be found in Appendix A.

Table 1
Summary of the economic sector representation¹²³ in ⁴SATIM and their⁵⁶⁷ main drivers.

Sectors	Subsector disaggregation	End use disaggregation	Useful energy demand drivers
Agriculture	None	Irrigation, heating, processing, traction, other.	Sectoral GDP ⁵
Residential	High, medium and low-income households	Cooking, water heating, space heating and cooling, refrigeration, lighting, other electric usage	Population, ⁶ Household size, ⁷ Household income, ⁸ electrification rate ⁹
Commercial	None	Cooling, space heating, cooking, lighting, refrigeration, water heating, public lights, public water	Sectoral GDP ⁵ , building stock floor area ¹⁰
Industrial	Iron and Steel, Ferroalloys, Non-metallic minerals, Platinum Group Metals, Aluminium, and Pulp and Paper mining, chemicals, food beverages and tobacco, precious, and non-ferrous metals, and general manufacturing	Tonnes produced by Industrial processes	Sectoral GDP ⁵
		Boiler and process heating, cooling, HVAC, lighting, fans, pumping, compressed air, electrochemical and other electrical services	Sectoral GDP ⁵
Transport	Air, Freight and Pipeline	Freight tonne km by rail and road (1 light vehicle class, 9 heavy vehicle classes)	Transport GDP ⁵
	Private passenger	Passenger kilometre travel by Cars, SUV, motorbikes	Household income, population ⁶ , vehicle ownership ¹¹ etc

⁴ The higher temporal resolution model allows more robust representation of demand and electricity generation from technologies such as intermittent renewable energy and storage but can increase the time needed for each model run considerably in particular in the linked SATIMGE model where each run requires several iterative runs of both SATIM and SAGE.

4.2. Overcoming data limitations

Data limitations, in terms of availability and in terms of quality of data is a common challenge faced by developing countries. Several approaches have been used in the development and maintenance of SATIM to overcome these challenges. Where official data is incomplete or inconsistent, SATIM integrates and reconciles data from multiple sources, including industry reports and grey literature. For areas with limited data, such as informal industries or rural energy use, SATIM employs proxy indicators and scaling factors based on better-documented sectors or data from comparable countries. This approach is similar to that used by other researchers in South Africa and in other developing countries; for instance, Howells et al. [19] describe the application of surveys and tools of modern energy modelling to estimate rural energy demand in South African villages, while Bhattacharyya and Timilsina [20] discuss the use of urban proxies and other techniques for modelling energy demand in developing countries. In the industrial sector, where detailed process data is sometimes lacking, SATIM uses benchmarking techniques and international best practice data, an approach also employed in China's TIMES model Chen et al. [21]. Lastly, for long-term projections, where historical data may be insufficient, SATIM incorporates expert judgement and stakeholder inputs, a method also used in Brazil's energy modelling efforts Lucena et al. [22].

SATIM's flexible model year and temporal resolution structure also allow the model to be adjusted for different research questions according to data availability. As better data is collected, the model can be updated and recalibrated, as needed.

Lastly, as identified in Bhattacharyya and Timilsina [20] SATIM delivers richer and more reliable modelling results due to the institutionalised modelling capacity that exists in the country. The continuous process of updating and recalibrating the model by a dedicated team as new data and information become available is a key requirement for maintaining a model capable of input into policy formulation.

4.3. Representing complex socio-economic factors

The energy system at a household and community level is highly complex, due in part to the significant development imperatives faced in South Africa. The triple challenge of deep set and pervasive poverty, unemployment and inequality, requires significant effort to capture these important dynamics within an energy model.

The residential sector in SATIM captures energy used by households to supply household energy services. Energy service levels and fuel choice are driven by population and income [23,24]. In South Africa and other developing countries, energy use patterns are complex and the welfare impacts of fuel switching are a key developmental goal for energy policy [25]. Households in SATIM are represented in three income groups, to capture the influence of income on energy service demands, appliance ownership and fuel choice. Households in the lower income group are more likely to be unelectrified, own significantly fewer appliances, and are more likely to supply thermal energy needs with coal, wood or paraffin [26–28]. Each household grouping in the model is distinct in terms of the level of useful energy demand assigned to energy services and the fuel and efficiency of technologies that supply energy services. A further advantage of representing households in income

groups is that the model can incorporate and react to policy interventions which target energy poverty.

The share of households allocated to each income group in the base year relies on Stats SA data, and specifically the national population census [34] and community survey [35]. Over the modelling horizon, the income band of each income group remains static and an increase in household income is captured as a movement into a higher income group, increasing the share of households in the higher income groups over time. By modelling households in this way, the increase in energy service demands in the model is consistent with household income levels (and vice versa).

The base year calibration, in terms of fuel use and energy service demands, relies on a combination of data from large national surveys to capture fuel choices in income groups and appliance ownership levels, and bottom-up surveys that capture user behaviours and the quantity of energy used by households (see for example [24] for further detail on the data sources and energy service demands assumed). The demand for energy services in each group is therefore an indicative estimate and the representation of the sector in further detail is limited by data availability.

The base year calibration of UE attributed to each household group is calculated as follows:

$$U_{ijn} = EI_{ijn} \times H_{in}$$

Where,

U_{ijn} is the useful energy demand for income group i using energy service j in year n (GJ),

EI_{ijn} is the energy Intensity for income group i using energy service j in year n (GJ/per household/year) and,

H_{in} is the number of households in year n (households).

Within each income group, the degree to which households can switch between fuels is controlled by exogenous assumptions which stipulate an upper and lower share that each fuel-based technology can supply cooking, water heating or space heating to households. Under this configuration households can switch to alternative, cheaper fuels, or fuels with lower GHG emissions, but only to the degree that the exogenous assumptions allow.

4.4. Modular structure and flexibility

The TIMES framework is designed to allow for interconnected, but unique modules representing different sectors of the energy system. The SATIM model leverages this feature, allowing the addition of new sectors and sub-sectors as more data becomes available and in response to different research questions. For example, SATIM's transport sector was significantly adapted following the development of a vehicle parc model and quantification of demand for passenger and freight transport [29]. This adaptation was possible without necessitating changes to other sectors. Similarly, new technologies and policies can be incorporated as needed. This is further explained in Section 6.

The model has been regularly maintained and enhanced for almost two decades. This has allowed the detail within each sector to develop over time. The level of detail in which each sector is modelled is often a reflection of the relative contribution of the sector to total consumption, but it is also a reflection of the level of funding that has been available for developing a sector in the model. For example, the Transport sector, which is a large energy user and has been a focus of model development for several years, has a high level of detail compared to the Agricultural sector, which accounts for a relatively small percentage of TFC and emissions and is quite simplistically represented in SATIM. Furthermore, the modular structure enables SATIM to be easily linked with other models, such as the CGE model for economy-wide analysis, as discussed in section 5.5.

⁵ GDP all sectors: (base year GDP from the South African Social Accounting Matrix (SAM) [65], thereafter SATIMGE).

⁹ Electrification rate is assumed based on historic trends and targets.

¹⁰ Floor area [63].

¹¹ Vehicle ownership (eNatis) [48].

⁶ Population and population growth [1,66].

⁷ Household size (calculated using [67]).

⁸ Household income (base year household income from Ref. [65], thereafter SATIMGE).

4.5. Accounting for GHG emissions

SATIM models all GHG emissions emitted by combustion and industrial process emissions classified using the IPCC's 2006 guidelines and where applicable the 2019 Refinement, to match South Africa's national GHG inventory and NDC target. SATIM also models local air pollutants from all combustion activities (NO_x, SO_x, PM₁₀), and water use. South Africa's GHG emissions are highly concentrated in a small number of facilities, including the existing coal fleet, a synthetic fuels plant, and smaller emitters in iron and steel, ferrochrome and cement. Large sources are modelled at facility level, and emissions estimates for the whole energy system are frequently calibrated against South Africa's national GHG inventory. This level of detail equips SATIM to be used in an integrated GHG emissions modelling framework (SATIMGE) to address national policy challenges such as undertaking technical analyses to support the setting of NDC targets.

4.6. Integration with other models

An important and novel expansion of SATIM has been the hard-linking of SATIM to the energy-enhanced South African General Equilibrium (eSAGE) model [30] and to two spreadsheet-based models to model GHG emissions from IPCC sectors AFOLU and waste, forming SATIMGE. eSAGE is a dynamic recursive, economy-wide, Computable General Equilibrium (CGE) model of South Africa which has a high level of granularity in the energy sector to allow the model to capture the price and investment effects of technology change in this sector more accurately. The linked SATIMGE model captures the technical detail required for full energy systems modelling, as well as the economic detail and responsiveness for assessing the impact of investments and costs on economic activity across the economy. The link is described in detail in Ref. [31].

SATIMGE can capture several important energy-economy interactions dynamically. In each iteration of the linked model, energy service needs in SATIM are driven by economic activity by sector or subsector in eSAGE. The models are run recursively: the energy system in SATIM is optimised to meet the revised energy service needs subject to any technology/policy constraints applied, the resulting energy supply investments and demand composition in SATIM is passed to eSAGE and the models run iteratively until convergence between the GVA modelled in eSAGE and the GVA demand drivers in SATIM.

A strength of the linked model is that eSAGE provides SATIM with an internally consistent projection of the economy (sectoral economic activity and household income) driven either directly (e.g. commerce) or indirectly (e.g. residential) by sectoral changes in GDP. For example, GDP increases in manufacturing in eSAGE increase energy service needs in that sector in SATIM. Increases in manufacturing in eSAGE also drive employment at defined skills levels which results in a growth in household income levels which drives energy services in the residential sector and transport sectors in SATIM. Rises in household income level are therefore consistent with labour and skills levels and economic growth, allowing the linked model to capture options such as the welfare benefits of localisation of RE components endogenously.

Adding AFOLU and waste components to the SATIMGE modelling framework provides the basis for modelling economy-wide emissions pathways (for instance, to reach net zero CO₂ in the medium- or long-term) which are technically and economically consistent and capture the impact of non-energy sources and sinks of GHG emissions.

5. Applications

SATIM has been applied to a wide range of research questions. The hard linking of SATIM and eSAGE has allowed the expansion of research to areas such as just transitions and economically and climate optimal development pathways, where reporting on economic indicators that are of interest to policy makers such as employment and household

welfare is imperative. A selection of applications, that are of specific interest to South Africa, are briefly described in this section to demonstrate the breadth of analysis that is possible with SATIM. The studies mentioned below focus on coal, GHG emissions mitigation and uncertainties. Coal is the dominant source of primary energy in South Africa, and largely due to this, South Africa is the Continent's largest GHG emitter.

SATIM has been used to analyse the implications of coal transitions [32], uncertainties in the coal price [33,34] and, as much of South Africa is arid and water scarcity can impact the long-term viability of power plants, the implications of water constraints for power sector expansion in South Africa [35]. In 2022, SATIM was used as an evidence base to support the process of developing South Africa's Just Energy Transition Investment Plan (JET IP) [32]. The analysis and model results provided a sounding board for different scenarios, and critical cost and investment figures to underpin the underlying calculations in the JET IP. To develop insights into key uncertainties in long-term energy planning and the associated emissions in South Africa, SATIM was used to generate probabilistic forecasts of GHG emissions using monte-Carlo generated projections of key variables [33]. This modelling was valuable in shedding light on the sensitivity of emissions pathways to changes in these key variables. A version of SATIM which tracks water use by power plants has been used to highlight the implications of water constraints for energy planning in South Africa [35]. The Thirsty Energy: Water-Smart Energy Planning in South Africa study tracks water use for wet and dry coal-fired power plants at the individual power plant level. The study points to the importance of including nexus trade-offs between water use by power plants and other sectors in expansion plans in South Africa.

In 2022, the World Bank launched the Country Climate and Development Reports (CCDRs). These reports, which assess the intersections between climate and development objectives, are compiled for several countries. For South Africa's CCDD [36], SATIMGE was used to analyse the energy sector, industry and macroeconomy impacts of different pathways to Net Zero. The analysis also included different finance and green industry scenarios to compare the effect of government policy on socio-economic outcomes. The energy sector findings highlighted how SA could leverage its low-carbon efforts to advance its development agenda by adopting the right mix of policies and strategies and harnessing external sources of finance. SATIMGE has also been used to provide a first pass at analysing 2050 net zero pathways for South Africa [37] for the Presidential Climate Commission (PCC). Here the focus of the analysis was on estimating South Africa's fair share, looking at different carbon budgets and analysing a comprehensive set of potential pathways to NZ in 2050. The complex nature of this analysis led to the recommendation for a deeper study into NZ pathways for South Africa, which is currently ongoing.

SATIM has also been used to provide an evidence base in the process of updating South Africa's mitigation targets in its First Nationally Determined Contribution [38]. The model was used to unpack the feasibility of achieving, and potentially increasing the ambition of South Africa's NDC given updated growth rates and technology advances.

SATIM has also been applied to explore an alternative integrated resource plan for South Africa [13]. The analysis had three main objectives: understanding key drivers of electricity system; addressing the claim that the official Integrated Resource Plan (IRP 2019) [39] presented least-cost options for the electricity sector; and lastly to provide additional insight into what would be required for a more ambitious decarbonisation pathway for the sector in the light of the requirements of the Paris Agreement.

6. Discussion and areas of future work

SATIM captures key components and novel aspects of South Africa's entire energy system and has been built and maintained over two decades despite significant data and capacity constraints. The model

allows for integrated analysis of complex cross-sectoral policy objectives and sector-specific policies, as well as deep dive quantitative analysis of specific technology or sub-sector change. The hard link with a CGE model enables demand responsiveness to prices and provides consistent economic structure representation.

To remain relevant, SATIM is continuously being improved and updated as new data and information become available, and as new research questions arise. Some of the main areas for future research are listed below.

- **Spatial representation:** Currently, SATIM represents South Africa's energy system as a single node. Future iterations should aim to include multiple regions to better capture the spatial diversity of supply, demand, and resources. This would improve the model's ability to optimise the energy system and account for factors like transmission costs and grid infrastructure investments.
- **Capturing household preferences:** The model could benefit from incorporating more detailed choice models for the residential demand sector. This would help capture the heterogeneity of household preferences beyond just cost considerations. Integration with other models: Further research could explore the benefits of soft-linking SATIM with more detailed choice models to improve its representation of consumer behaviour.
- **Transport sector improvements:** Modal choice is currently exogenously defined. Future research should focus on disaggregating transport demand for major cities to reflect their unique characteristics, such as available public transport options and road congestion levels.
- **Regional trade dynamics:** Regional trade is likely to expand in future and having the flexibility to include demand and supply options in neighbouring countries explicitly, at a similar level of detail to SATIM, will allow exploration of optimal energy pathways for the region.
- **Temporal resolution:** While the model can accommodate different temporal resolutions, further research could explore the implications of various timeslice resolutions on RE integration and model run time.
- **Sector-specific enhancements:** Continued development of less detailed sectors, such as agriculture, could provide a more comprehensive representation of the entire energy system.

7. Conclusion

This paper has outlined some of the challenges and opportunities of energy systems modelling in developing countries using the experience of developing and maintaining the South African TIMES Model (SATIM) over almost two decades. South Africa is an interesting case due to its complex socio-economic landscape, characterised by a triple development challenge of high levels of inequality, poverty, and unemployment coupled with a carbon-intensive economy. A complex modelling approach has been used to capture these intricacies while overcoming significant data limitations and human capacity constraints.

Upon reflection, key to the successful development of SATIM has been: the sustained improvement of the data and structure of the model over decades; the integration of SATIM with other models; the specificity of the model structure to depict South Africa's specific energy system and challenges, and; the development of the model in a way that allows for modular improvements allowing, for example, the time slice resolution or expansion of the model structure focussed individual sectors and subsectors to be more easily accommodated.

The integration of models such as a transport sector vehicle parc model, and a residential sector stock model has allowed granular sectoral detail and dynamics to be accounted for in SATIM. In particular the integration of SATIM with an economy-wide model, which provides a consistent framework through which complex energy-economy interactions can be analysed including the socio-economic implications of

scenarios, critical policy interventions and the JET more broadly.

The model's continuous refinement has allowed researchers to address a wide range of critical policy issues, from providing an evidence base for South Africa's Nationally Determined Contribution targets to analysing just transition pathways and exploring feasible net-zero scenarios. The applications of SATIM underscore its importance in providing evidence-based insights for South Africa's energy and climate policy decisions. As the country navigates ongoing energy transitions and challenges, including the need to decarbonise while addressing development imperatives, SATIM's role in informing policy remains crucial.

However, due to the changing socio-economic, technology and policy landscape, the work of improving and expanding SATIM is ongoing. Future developments, such as enhancing spatial representation, improving transport sector representation, and incorporating more detailed regional trade dynamics, will further increase the model's utility and accuracy.

The experience of developing and maintaining SATIM offers valuable lessons for other developing country teams facing similar challenges in energy systems modelling. It highlights the importance of long-term, sustained modelling efforts in building capacity and providing consistent, reliable analysis for policy and decision makers. These efforts are essential, and increasingly so, in navigating the complex interplay between energy systems, economic development, and climate change mitigation and require sustained support at both the local and international level.

In conclusion, SATIM stands as a testament to the value of comprehensive, adaptable energy systems model built and maintained within developing country contexts. As South Africa and other countries continue to navigate energy transitions, while also striving for sustainable development, tools like SATIM will play an increasingly vital role in providing an evidence base for decision makers as they chart a path toward a more sustainable and equitable energy future.

CRedit authorship contribution statement

Alison Hughes: Conceptualization, Visualization, Writing – original draft, Writing – review & editing. **Tara Caetano:** Conceptualization, Visualization, Writing – original draft, Writing – review & editing. **Guy Cunliffe:** Writing – review & editing. **Faaiqa Hartley:** Conceptualization, Writing – original draft. **Andrew Marquard:** Writing – review & editing. **Bryce McCall:** Conceptualization, Visualization, Writing – original draft. **Bruno Merven:** Conceptualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Alison Hughes reports financial support was provided by South African National Energy Development Institute (SANEDI). If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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data, and the soft linking of the energy-economy-wide model, SATIM-GE.

APPENDIX A

A.1 SATIM energy balance data sources

Table A-1 provides an overview of data sources used to calibrate a base year energy balance for SATIM.

Table A-1
SATIM Energy Balance Data Sources

Sector	Data Source
All fuels	- DOE Energy Balance 2017 [17] - Stats SA Supply and Use Table (SUT) 2017 [40] - DFFE GHG inventory 2017 [41] - NBI Net Zero studies [42]
Electricity	- Eskom electricity sales (Eskom annual reports e.g. Ref. [43] and personal comms) - Eskom coal purchases (Eskom annual reports e.g. Ref. [43])
Gas	- Sasol gas sales and purchases [44] - Sasol Annual GHG Emissions Report SSO_FY18 [45]
Coal	- PETRO SA Integrated report 2020 [46] - South African Mining Industry 2017 [47] - Personal communication with experts in the coal industry
Liquid fuels	- DMRE liquid fuel sales (pers. comms) - NAAMSA vehicle sales (purchased dataset) - eNATIS vehicle park estimates [48] - Calibration explained [49]
Biomass	- Pulp and paper and sugar reports (e.g. The South African sugar industry directory 2016/2017 [50], SAPPI 2022 Group Sustainability Report [51]) - Residential – estimated based on [52–56] amongst others

A.2 Further information on SATIM supply sectors

This section provides further detail on the structure and methodology applied in the supply sectors in SATIM. Cost and technology parameter assumptions are updated regularly, recent examples of the parameterisation of the supply sector in terms of fuel costs and technology costs and characteristics can be found in Refs. [57,58].

A.2.1 resource extraction

Existing oil and gas fields are largely depleted. New gas fields are only included in specific scenarios, as extraction costs are highly uncertain. The types of biomass currently included at primary level are bagasse (to sugar industry), Biomass for Pulp (to Pulp and Paper), Wood (to bio feedstock production, households and industry), and Other Biomass (includes biogas) to the electricity sector.

A.2.2 trade

Import volumes are endogenous and are based on scenario-based projected international commodity prices. Imported liquid fuels derived from crude oil are set without taxes, and are indexed to the price of crude oil. Exports for energy commodities are specified exogenously based on historical levels.

A.2.3 power sector

New build options include potential imports from SAPP (by project) and nuclear, wind, solar PV (grid and rooftop), solar thermal, coal, diesel and gas (OCGT and CCGT), hydro and pumped storage plants as well as battery technologies. In addition to technology and cost parameters, all new build options are subject to constraints which limit either the cumulative capacity, annual maximum investments in new capacity, or the growth in new capacity additions. This is done to reflect physical limitations such as port limitations, local skills capacity and manufacturing capacity. Costs and other parameters of both new and current power plants are regularly updated.

$$\sum_i (CC_i \times \text{Installed Capacity}_i) \geq (RM + 1) \times \text{PeakDemand (grid)}$$

Where CC_i = Capacity Credit of grid-based electricity generation technology i and RM is the exogenously specified reserve Margin, PeakDemand is the peak demand seen by the grid.

In SATIM, solar PV, solar thermal plants without storage and wind are by default conservatively assumed to be unable to contribute to the reserve margin (i.e. $CC = 0$).

New transmission infrastructure has been identified by policy makers as the most important constraint to the rollout of further RE infrastructure. SATIM is currently a single-node model and therefore cannot account for current or worsening transmission constraints in parts of the network as new build occurs. To help with this limitation, new capacity (beyond what is already committed until 2030 when it is assumed the bottlenecks are alleviated) is assigned a lower resource quality. Transmission expansion costs are modelled in two steps: Step 1 tracks peak demand, and has a higher investment cost per kW; Step 2 tracks the sum of installed capacity minus Step 1 capacity, and has lower cost. Step 2 accounts for grid expansions that are required to accommodate spatially diffuse installations of generation that extends beyond peak demand.

Transmission losses are accounted for as an aggregate average loss on the centralised electricity supply. Distribution losses and costs are accounted for separately by sector. Residential and agricultural losses are assumed to be higher than for the commercial sector and light manufacturing, and lower for heavy manufacturing, which in many instances is connected directly to the transmission network. Assessment of investment and maintenance costs follow the same logic. Losses in the transmission of electricity to and from these technologies is accounted for as follows.

- Cogeneration occurs after transmission and electricity is fed into the distribution grid to meet industrial electricity demand. The heat generated is used to meet industrial heat demand.
- Distributed Generation has expanded rapidly in recent years due to cost reduction of PV and grid supply interruptions across all sectors. Distributed energy technologies built within a sector such as embedded solar or storage are not subjected to distribution losses and do not feed electricity into the central network to reach the other sectors.

A.2.4 Liquid fuels

Commodities produced by the South African liquid fuels sector at present are petrol (gasoline), diesel, kerosene (jet fuel/illuminating paraffin), aviation gasoline, liquid petroleum gas (LPG), heavy fuel oil (HFO) and methane-rich gas (produced by the synthetic fuels industry only). In SATIM, liquid fuels can be sourced from crude oil refineries, synthetic fuels plants (manufactured from coal and gas), and from imports. Product slates are modelled by applying an assumed limits on the shares for each output commodity. In existing refineries, the product slate is fixed over the planning period (based on the historical production of each refinery), and new refineries are modelled in a way that can accommodate a varying product slate.

A.2.5 Hydrogen

Hydrogen can be produced in SATIM either by coal gasification, via steam methane reforming (SMR) or via electrolyzers. Two types of electrolyzers are currently considered: polymer electrolyte membrane (PEM) electrolyzers, and alkaline electrolyzers, which differ in cost assumptions, flexibility (high temporal resolution configurations), and that PEM drives an additional demand for the Platinum sector in the model.

A.2.6 Bio feedstocks

The only bio feedstock currently considered in ‘transformation’ is biochar production needed for the decarbonisation of the ferrochrome sector. Liquid biofuels (such as methanol and bio-diesel), are excluded from the current version of the model, as they can’t compete with other alternatives given the water constraints faced by South Africa.

A.3 Further information on SATIM demand sectors

A.3.1 Residential sector

In SATIM, there is only one appliance type for each fuel supplying each energy service in each income group apart from water heating. Changes in the efficiency at which a fuel supplies energy services in the model therefore captures the change in the average efficiency or average energy intensity at which energy services are delivered by a particular fuel. The choice to represent appliances as an aggregate is due to the diverse nature of technology and cost choices for even simple appliances, and also due to the wide variety of influences and preferences, in addition to cost, that play a role in technology choice in the sector.

Geysers and SWHs or heat pumps are represented separately due to the dominance of water heating demand, the large opportunity for improving technology efficiency and the targeting of water heating for at least 50 % electricity saving in new buildings in the national SANS10400 standard [59, 60].

A.3.2 Transport

The transport sector in SATIM includes energy used for passenger and freight transport by road and rail, in pipeline transfers and in aviation. Energy services demand for passenger and freight transport is expressed in passenger kilometres (p.km) and tonne kilometres (tkm). Final energy consumption by the transport sector is driven by technology choice (endogenous) and other assumptions such as vehicle occupancy rates. Conversion efficiency (final energy to energy services) differs between vehicle types, fuel type and the age of the vehicle parc and to some degree the patterns of utilisation of different vehicle types. The energy service demand in SATIM is defined in terms of passenger kilometres (pkm) and tonne kilometres (tkm).

A calibrated vehicle parc model is used to estimate demand for passenger kilometres (pkm) as well as the parameters of the existing vehicle parc (as discussed in Ref. [61]). Aspects of transport included in the parc model are the size of the existing vehicle fleet, annual vehicle sales, annual vehicle scrapping, distance travelled per vehicle, fuel sales and vehicle fuel efficiency. Outputs of the vehicle parc model are total kilometres travelled, the average age of vehicles in the vehicle fleet and the average efficiency of the vehicle fleet. These components allow efficiency or intensity of transport to change with vehicle stock changes and an increase or decrease in vehicle ownership in response to population and income changes. Certain factors affecting the distance travelled and fuel efficiency, for instance traffic congestion, are difficult to quantify as they are not well understood. The annual distance travelled by vehicles, is translated into a demand for p.km by assuming average occupancy rates for the different vehicle types in SATIM. A recent update of the vehicle parc model calibration can be found in Ref. [49].

The demand for pkm is split between income groups and a category ‘other’ to account for commercially- and government-owned cars. The passenger demand projection model (in excel) uses assumptions around private vehicle ownership by income group, vehicle mileage, vehicle occupancy, public mode shares, average mode speeds, and a travel time budget, to derive vehicle-km demand by passenger vehicle class for households. This is combined with a transport-GDP linked projection of the non-household owned cars to give a total passenger vehicle-km demand projection for road vehicles. The passenger-km projections for rail are derived from assumptions around future mode shares i.e., the degree to which passengers are using private or public transport. The transport model in SATIM is thus implemented as depicted in Figure A-1.

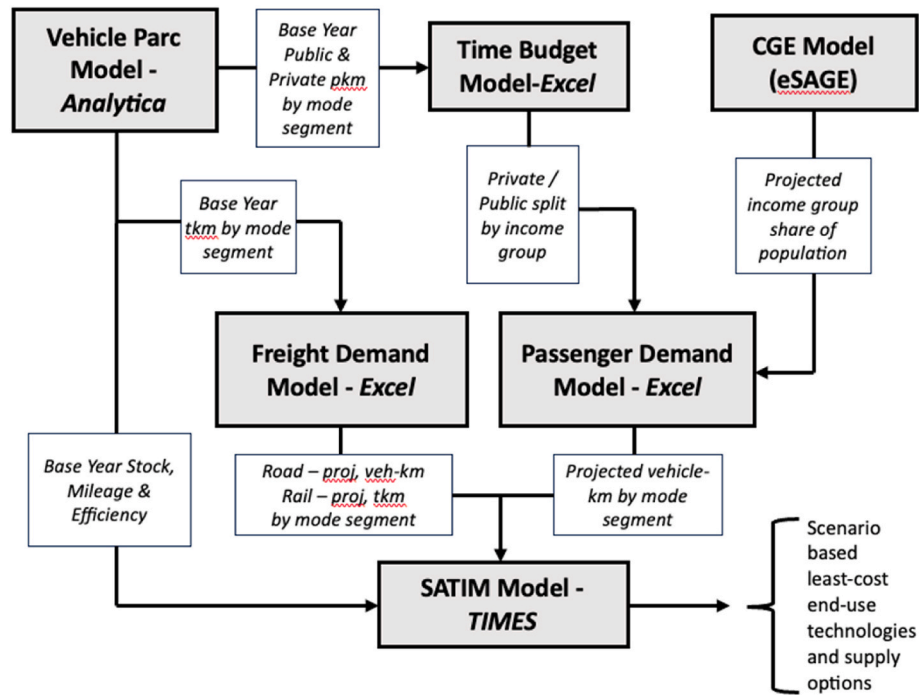


Figure A-1. An overview of the SATIM transport sector model [37]

Demand for vehicle-kilometres of freight in different vehicle classes is linked to sectoral GDP growth, applying load factors and modal shares, which influence fuel efficiency.

Within vehicle classes the model is able to optimise within a constraint on total sales of new vehicles. Scrappage factors are applied that reduce the existing vehicle fleet, new vehicles sales are therefore needed to replace older vehicles as they are scrapped as well as to meet new demand.

A.3.3 Commerce

The commercial sector includes energy used in private and public sector buildings as well as public lighting and water treatment plants. The commercial sector is a growing sector in South Africa, in terms of both its contribution to GDP and the amount of building floor area. Energy consumption in this sector grows in relation to floor area. Growth in floor area over the model horizon is driven by GDP at a ratio of 0.7:1.

Demand for energy services is based on the energy intensity of energy services needed for each unit of floor area in different building types (PJ/m²). Floor area in SATIM is accounted for as both new and existing building floor area. New commercial floor space is assumed to be more energy efficient, having more options to adopt energy efficient technologies as well as a lower demand for energy services due to improvements in building design. New floor area in the model is disaggregated into building category types which capture changes in building efficiency over time.

There is very little information available to estimate the quantity of electricity or other fuels supplying individual energy services in the commercial sector. The estimated intensity (MJ/m²) of supplying energy services draws on the 2012 USA Commercial Buildings Energy Consumption Survey (CBECS) [62]. A weighted sum of the average electricity intensity of energy services for all CBECS building types located in the climate zone most similar to South Africa (i.e., climate zone <4000HDD & <2000CDD) is used. The weighting, by floor area type, is estimated by summing floor area reported by Stats SA from 1990 [63] and stock estimates from Ref. [64].

A.3.4 Industry

In SATIM, two methodologies are applied to model industrial energy consumption. The first relies on estimates of energy service needs (heating, lighting, cooling, compressors etc) and is generally applied to industries that have lower energy intensities, a variety of products and are more numerous. The second utilises an estimate of the energy intensity of industrial technology processes and is typically applied to sectors where products are more uniform and the energy intensity of production is high – for example steel works or cement kilns. The two approaches are described in more detail below.

Approach one is applied to the mining, chemicals, food and beverages, precious and non-ferrous metals (excluding Platinum group metals, and aluminium) and a variety of other manufacturing. In this approach useful energy services, linked to economic activity, drive energy consumption. The useful energy service requirement (e.g. process heat, compressed air, etc) in each sub-sector is estimated based on an estimate of final energy supplying each energy service and the efficiency at which energy services are met. This is demonstrated in Figure A-2, where the level of useful energy services needed for process heating, and the efficiency of the boiler, determines the amount of final energy (coal) consumed.

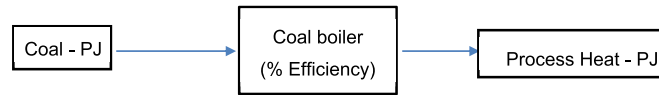


Figure A-2. Technology representation in approach one subsectors in industry - example of a boiler technology for process heat

The estimate of base year demand for useful energy services $U_{i,T,f,u}$ is calculated for each industry subsector i , technology T , fuel or commodity f , and end-use service u (heating, lighting etc.) using the equation below:

$$U_{i,T,f,u} = \eta_{i,T,f,u} \times F_{i,T,f,u}$$

Where $\eta_{i,T,f,u}$ is the technology efficiency, and the final energy consumed F to supply the energy service is calculated according to the share of technologies supplying the energy service

$$F_{i,T,f,u} = S_{i,T,f} \times U'_{f,i,u} \times E_{f,i}$$

where,

$E_{f,i}$ is total final consumption of fuel f by subsector i (Joules), and

$U'_{f,i,u}$ is the % share of final energy supplying end use service u

$S_{i,T,f}$ is the % share of technology T assumed to supply the end use service

Assumptions around technology efficiency and energy service levels are drawn from literature and from energy audits and surveys.

Approach two is used for the Iron and Steel, Ferroalloys (Ferro-Chrome, Ferro-Silicon, Ferro-Manganese, and other Manganese alloys in South Africa), Non-metallic minerals (which includes the production of cement, glass, lime, and bricks), aluminium, and pulp and paper sectors, and Platinum Group Metals. The demand for final energy in these sectors is calculated endogenously based on the energy intensities specific to technology processes and their level of production. For example, in Figure A-3, the demand for coal and coke by blast furnaces in the production of iron is calculated based on the technology specific energy intensity (GJ/t) of iron production in South African blast furnaces. In order to apply approach two, the share of production by technology type and the energy intensity of production in South Africa must be known or estimated.

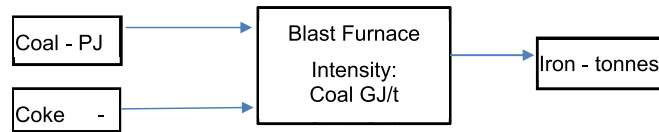


Figure A-3. Example of technology representation in approach two subsectors in industry

The production output O by product P is given as:

$$O_P = \sum_T A_{T,P}$$

where $A_{T,P}$ is the activity of a technology T producing product P .

Final energy consumption F of fuel f for an industry is then:

$$F_f = \sum_T (A_{T,P} \times I_{T,f})$$

where $I_{T,f}$ is the energy intensity for technology T , and f represents the fuels.

Figure A-4 shows how existing technology capacity (blue) and new technology additions (green) are explicitly represented in the model in the steel, cement, Ferro-Alloy and pulp and paper sectors. Existing technologies and plants can be retrofitted to improve efficiencies. New technologies or technologies using alternative fuels can be introduced to improve the efficiency of production.

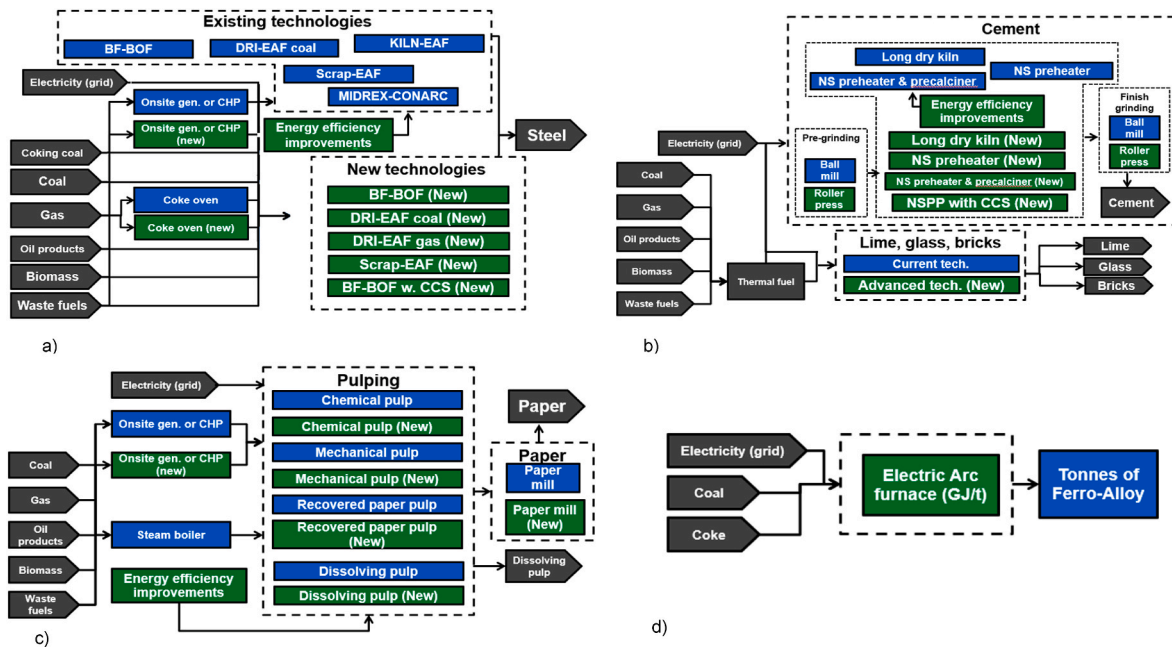


Figure A-4. Structure of the a) Iron and Steel b) Cement c) Pulp and Paper industry and d) Ferroalloys representation in SATIM

Data availability

Data will be made available on request.

References

- [1] United Nations Department of Economic and Social Affairs Population Division vol. 2022, World Population Prospects, 2022.
- [2] ADB, African Economic Outlook 2024: Driving Africa's Transformation: Country Notes, African Development Bank Group, 2024.
- [3] World Bank Group, World Development Indicators, GDP growth (annual %). <https://databank.org/reports.aspx?source=2&country=ZAF#>, 2024. (Accessed 22 November 2024).
- [4] National Planning Commission, NPC economy series Energy, 2019, p. 25, version 1.0.
- [5] DMRE, Energy Statistics, Energy balances, 2021 commod. Flows Energy Balanc, 2021. https://www.energy.gov.za/files/energyStats_frame.html.
- [6] Y. Ye, S.F. Koch, Measuring energy poverty in South Africa based on household required energy consumption, *Energy Econ.* 103 (2021) 105553, <https://doi.org/10.1016/j.eneco.2021.105553>.
- [7] P. Lopion, P. Markewitz, M. Robinius, D. Stolten, A review of current challenges and trends in energy systems modeling, *Renew. Sustain. Energy Rev.* 96 (2018) 156–166, <https://doi.org/10.1016/j.rser.2018.07.045>.
- [8] S. Jebaraj, S. Iniyan, A review of energy models, *Renew. Sustain. Energy Rev.* 10 (2006) 281–311, <https://doi.org/10.1016/j.rser.2004.09.004>.
- [9] N. Neshat, M.R. Amin-Naseri, F. Danesh, Energy models : methods and characteristics, *J. Energy South Afr.* 25 (2014) 101–111.
- [10] R.B. Hiremath, S. Shikha, N.H. Ravindranath, Decentralized energy planning ; modeling and application — a review, *Renew. Sustain. Energy Rev.* 11 (2007) 729–752, <https://doi.org/10.1016/j.rser.2005.07.005>.
- [11] P. Laha, B. Chakraborty, Energy model – a tool for preventing energy dysfunction, *Renew. Sustain. Energy Rev.* 73 (2017) 95–114, <https://doi.org/10.1016/j.rser.2017.01.106>.
- [12] J.F. De Carolis, K. Hunter, S. Sreepathi, The case for repeatable analysis with energy economy optimization models, *Energy Econ.* 34 (2012) 1845–1853, <https://doi.org/10.1016/j.eneco.2012.07.004>.
- [13] B. McCall, J. Burton, A. Marquard, F. Hartley, F. Ahjum, G. Ireland, B. Merven, Least-cost integrated resource planning and cost-optimal climate change mitigation policy. Alternatives for the South African Electricity System, 2019.
- [14] A. Marquard, F. Ahjum, C. Bergh, H. Von Blottnitz, J. Burton, B. et, al Cohen, Exploring Net Zero Pathways for South Africa - an Initial Study, 2023.
- [15] Department of Energy, Integrated Energy Plan (Draft), 2016.
- [16] H. Winkler (Ed.), Long Term Mitigation Scenarios: Technical Report, 2007. Pretoria, South Africa.
- [17] DMRE, Energy Statistics, Energy balances, 2017 commod. Flows Energy Balanc, 2017. https://www.energy.gov.za/files/energyStats_frame.html. (Accessed 13 February 2024).
- [18] Eskom, Just Energy Transition (JET), Change Management Fact Sheet #001, 2021.
- [19] M.I. Howells, T. Alfstad, D.G. Victor, G. Goldstein, U. Remme, A model of household energy services in a low-income rural African village, *Energy Pol.* 33 (2005) 1833–1851, <https://doi.org/10.1016/j.enpol.2004.02.019>.
- [20] S.C. Bhattacharyya, G.R. Timilsina, Modelling energy demand of developing countries: are the specific features adequately captured? *Energy Pol.* 38 (2010) 1979–1990, <https://doi.org/10.1016/j.enpol.2009.11.079>.
- [21] W. Chen, X. Yin, H. Zhang, Towards low carbon development in China: a comparison of national and global models, *Clim. Change* 136 (2016) 95–108, <https://doi.org/10.1007/s10584-013-0937-7>.
- [22] A.F.P. Lucena, L. Clarke, R. Schaeffer, A. Szklo, P.R.R. Rochedo, L.P.P. Nogueira, K. Daenzer, A. Gurgel, A. Kitous, T. Kober, Climate policy scenarios in Brazil : a multi-model comparison for energy, *Energy Econ.* 56 (2016) 564–574, <https://doi.org/10.1016/j.eneco.2015.02.005>.
- [23] H. Winkler, O. Davidson, A. Kenny, G. Prasad, J. Nkomo, D. Sparks, M. Howells, T. Alfstad, Energy Policies for Sustainable Development in South Africa: Options for the Future, 2006.
- [24] A. Hughes, R. Larmour, Residential Electricity Consumption in South Africa Research Report, 2021. Cape Town, South Africa.
- [25] DOE, White Paper on the Energy Policy of the Republic of South Africa, 1998, pp. 1–110.
- [26] S.A. Stats, General Household Survey 2023, 2023.
- [27] S.A. Stats, Living Conditions Survey 2014/15, 2015.
- [28] DOE, A Survey of Energy-Related Behaviour and Perceptions in South Africa 2013, 2013.
- [29] B. Merven, A. Stone, A. Hughes, B. Cohen, Quantifying the Energy Needs of the Transport Sector for South Africa: A Bottom-Up Model, 2012.
- [30] B. Merven, C. Arndt, H. Winkler, The Development of a Linked Modelling Framework for Analysing the Socioeconomic Impacts of Energy and Climate Policies in South Africa, UNU-WIDER, Helsinki, Finland, 2017.
- [31] C. Arndt, R. Davies, S. Gabriel, K. Makrellov, B. Merven, F. Hartley, J. Thurlow, A sequential approach to integrated energy modeling in South Africa, *Appl. Energy* 161 (2016) 591–599, <https://doi.org/10.1016/j.apenergy.2015.06.053>.
- [32] J. Burton, T. Caetano, B. McCall, Coal Transitions in South Africa: Understanding the Implications of a 2°C-Compatible Coal Phase-Out for South Africa, 2018.
- [33] B. Merven, I. Durbach, B. McCall, Obtaining Long-Term Forecasts of the Key Drivers of Greenhouse Gas Emissions in South Africa, 2016.
- [34] I. Durbach, B. Merven, B. McCall, Environmental Modelling & Software Expert elicitation of autocorrelated time series with application to e3 (energy-environment-economic) forecasting models, *Environ. Model. Software* 88 (2017) 93–105, <https://doi.org/10.1016/j.envsoft.2016.11.007>.
- [35] World Bank Group, Modeling the Water-Energy Nexus: How Do Water Constraints Affect Energy Planning in South Africa, Washington DC, 2017.
- [36] World Bank Group (WB), South Africa Country Climate and Development Report, 2022. Washington DC.
- [37] A. Marquard, F. Ahjum, C. Bergh, H. von Blottnitz, J. Burton, B. Cohen, G. Cunliffe, A. Dane, F. Hartley, A. Hughes, G. Ireland, B. McCall, B. Merven, L. Stevens, H. Winkler, Exploring net zero pathways for South Africa - an initial study. <https://doi.org/10.25375/uct.22189150.v2>, 2023.

- [38] RSA, South Africa, First Nationally Determined Contribution under the Paris Agreement Updated September 2021, Tshwane, 2021.
- [39] Department of Mineral Resources and Energy, Integrated Resource Plan (IRP 2019), 2019. Pretoria, South Africa.
- [40] S.A. Stats, Supply and Use Tables : Revised 2018 and Preliminary 2019, 2022.
- [41] DFFE, National GHG Inventory Report: South Africa 2000 - 2020, 2022. Pretoria, South Africa.
- [42] NBI, South Africa `S Net-Zero Transition, 2022.
- [43] Eskom, Eskom Integrated Report 2018, 2018.
- [44] SASOL, Sasol Additional Analysts Information, 2017. https://www.sasol.com/sites/default/files/2022-08/AdditionalAnalystInformation-31December2017_0.pdf.
- [45] SASOL, Sasol Limited, Clim. Change 1 (2018) (2018) 1–71. <https://www.sasol.com/sites/default/files/2022-04/CDPSubmission2018.pdf>.
- [46] PetroSA, Integrated Annual Report 2020, 2020. <https://www.petrosa.co.za/PetroSAAR2020Final.pdf>.
- [47] Department of Mineral Resources and Energy, Sami 2016-2017, Resour. South African Miner. Ind, 2017. <https://www.dmre.gov.za/resources>.
- [48] National Traffic Information System, Live Vehicle Population, Stat. Live Veh. Popul. (n.d.).
- [49] A. Stone, B. Merven, T. Maseela, R. Moonsamy, Providing a foundation for road transport energy demand analysis: a vehicle parc model for South Africa, *J. Energy South Afr.* 29 (2018).
- [50] South African Sugar Association, South African Sugar Industry Directory 2016/2017, 2017.
- [51] Sappi, Group Sustainability Report, 2023. <https://cdn-s3.sappi.com/s3fs-public/2023-Sappi-Group-Sustainability-Report-Final.pdf>.
- [52] D.I. Banks, N.J. Grippin, C.M. Shackleton, S.E. Shackleton, J.M. Mavrandonis, W. R. Facility, S. Africa, Wood supply and demand around two rural settlements in a semi-arid Savanna, *S. Afr.* 11 (1996) 319–331.
- [53] M. Madubansi, C.M. Shackleton, Changes in fuelwood use and selection following electrification in the Bushbuckridge lowveld, South Africa, *J. Environ. Manag.* 83 (2007) 416–426, <https://doi.org/10.1016/j.jenvman.2006.03.014>.
- [54] D.B.K. Dovie, Rural economy and livelihoods from the non-timber forest products trade. Compromising sustainability in southern Africa? *Int. J. Sustain. Dev. World Ecol.* 10 (2003) 247–262, <https://doi.org/10.1080/13504500309469803>.
- [55] A. Williams, A. Eberhard, B. Dickinson, Synthesis Report of the Biomass Initiative, 1996.
- [56] N.C. Nkosi, S.J. Piketh, Solid fuel use in electrified low-income residential areas in South Africa : the case of KwaDela, Mpumalanga, *J. Energy South. Africa.* 32 (2021) 58–67.
- [57] RSA, South Africa's Just Energy Transition Investment Plan (JET IP), 2022. Pretoria, South Africa.
- [58] B. Merven, J. Burton, P. Lehmann-grube, Assessment of New Coal Generation Capacity Targets in South Africa's 2019 Integrated Resource Plan for Electricity, 2021.
- [59] SABS, South African National Standard SANS 10400-XA : 2021: the Application of the National Building Regulations Part X : Environmental Sustainability Part XA : Energy Usage in Buildings, 2021.
- [60] Government of South Africa, Draft Post-2015 National Energy Efficiency Strategy, 2016.
- [61] F. Ahjum, B. Merven, A. Stone, T. Caetano, Road transport vehicles in South Africa towards 2050: factors influencing technology choice and implications for fuel supply, *J. Energy South Afr.* 29 (2018), <https://doi.org/10.17159/2413-3051/2018/v29i3a5596>.
- [62] U.S. Energy Information Administration, Commercial building energy consumption survey (CBECS), 2012 CBECS surv. Data. <https://www.eia.gov/consumption/commercial/data/2012/>, 2012. (Accessed 31 July 2023).
- [63] S.A. Stats, Building Statistics P5041.1 (2024) 202405.
- [64] M. de Villiers, Greenhouse Gas Baseline and Mitigation Options for the Commercial Sector, 2000.
- [65] D. Van Seventer, R. Davies, A 2019 Social Accounting Matrix for South Africa with Occupational and Capital Stock Detail, 2023.
- [66] Statistics South Africa, Midyear Population Estimate 2019, Stat. South Africa, 2019, p. 24.
- [67] Stats SA, South African Census, Stats SA, Pretoria, 2011, 2012.